DeepFRI: Structure-Based Protein Function Prediction via Graph Convolutional Networks

GAHALAUT, Kovid MENG, Fan Rui WONG, Hok Fong
The Chinese University of Hong Kong
{1155173544, 1155194445, silaswhf}@link.cuhk.edu.hk

Abstract

In this report, we present a novel protein function prediction model, DeepFRI, that incorporates Long Short-Term Memory Language Models (LSTM-LM) and Graph Convolutional Networks (GCN) to predict protein functions based on residue-level features and protein structure. Results show that DeepFRI outperforms the sequence-only CNN DeepGo and the BLAST baseline in terms of prediction accuracy and generalization to unseen protein structures. The model is also able to identify key residues that contribute to protein function, providing insights into the molecular mechanisms underlying protein function. Hence, we will place emphasis on the model's formulation, experimental designs, and implications of DeepFRI.

1 Introduction

1.1 Challenges in Protein Function Prediction

The 3D structure of a protein is crucial for understanding its function, as it determines a wide range of functions from binding specificity and conferring mechanical stability, to catalysis of biochemical reactions, transport, and signal transduction. However, experimental determination of protein functions is time-consuming, labor intensive, expensive, and can only be applied to a small number of proteins. Currently, less than 1% of protein sequences have experimentally determined protein function information. To tackle this grand challenge in bioinformatics, many computational methods have been devised to automatically predict protein function.

1.2 Previous Works

Traditional machine learning classifiers, such as support vector machines, random forests, and logistic regression have been used extensively for protein function prediction. Scholars have established that multimodal approaches often outperform homology-based function transfer and that integration of multiple gene- and protein-network features typically outperform sequence-based features even though network features are often incomplete or unavailable.

Top-performing methods in the Critical Assessment of Function typically rely strongly on manually-engineered features constructed from either text, sequence, biological networks, or protein structure. However, these features are always unavailable for poorly studied organisms. Hence, it is important to develop methods that can predict protein functions based on sequence and structure alone.

In the last decade, deep learning with convolutional neural networks (CNNs) have been applied to protein function prediction, with models such as DeepGo that adopts 1D CNNs for recurring spatial patterns within sequence. Recent work has employed 3D CNNs to capture spatial information in protein structures, demonstrating the utility of structural features, but explicit 3D representations of protein structure at high resolution are not memory efficient.

2 Proposed Approach

Deep Functional Residue Identification (DeepFRI) is a novel deep learning model that integrates sequence and structure information for protein function prediction. DeepFRI combines Long Short-Term Memory Language Models (LSTM-LM) to capture residual-level features in protein sequences and Graph Convolutional Networks (GCN) to model residue interactions in protein structures. The model is trained end-to-end on a large dataset of protein sequences and structures with known functions from the Protein Data Bank (PDB) and homology models from SWISS-MODEL, and is evaluated on a held-out test set of unseen proteins.

2.1 Model Architecture

The first part of the model is an LSTM-LM *pretrained* on the entire Pfam database that takes a protein sequence as an input, and then processes the protein sequentially by *learning dependencies and patterns* within the protein sequence, such as motifs, repetitive structures, or specific amino acid arrangements. The LSTM-LM serves as a feature extractor, and is trained to predict an amino acid residue based on its context and position within a protein sequence. We analyze and suggest the three possible main advantages for choosing an LSTM model, featuring the following:

- *Local Context*: Information about neighboring amino acids.
- *Long-Range Dependencies*: Relationships between residues that are far apart in the sequence.
- *Biochemical Properties*: Implicit understanding of properties like hydrophobicity, charge, or propensity to form certain secondary structures based on sequence patterns.

The second part of the model is a Graph Convolutional Network (GCN) that accepts a *contact map* of the protein represented as a graph, where nodes represent residues and edges represent interactions between residues. The GCN learns to propagate residue-level information through graph convolutions that *convolve features structurally* rather than sequentially, as the key functional sites are often spatially close in the protein structure but distal in the sequence due to complex folds. It is highlighted that in the case of long protein sequences (e.g., > 400 residues), a CNN with reasonable filter lengths, would most likely fail to convolve over residues at different ends of the long sequence, even after applying multiple consecutive CNN layers. However, graph convolutions applied on contact maps are almost sufficient to access feature information from the complete structure in 3 layers or less.

Finally, the GCN protein representation of all layers are concatenated and is subsequently fed into a fully connected layer to produce protein function probabilities.

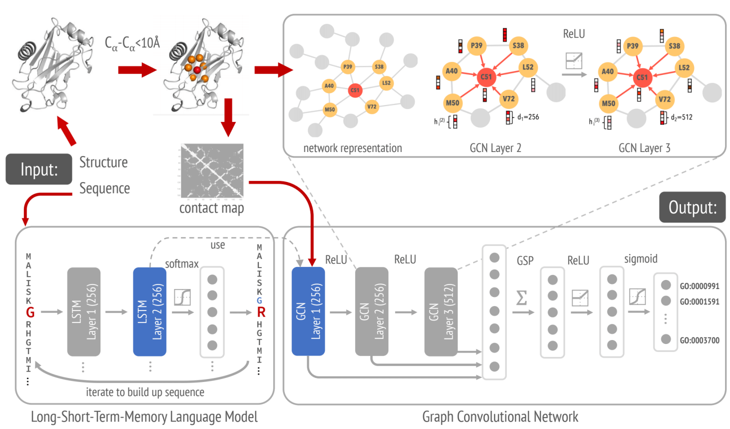

Figure 1: Architecture of DeepFRI.

Different models are trained to predict GO terms (one model for each branch of the GO: molecular function, cellular component, biological process) and EC numbers. The GO terms are selected to have at least 50 and not more than 5000 training examples, whereas EC numbers are selected from levels 3 and 4 of the EC tree as they are the most specific descriptors of the enzymatic functions.

2.2 Training Procedure

To train the GCN, experimentally determined and high-quality 3D atomic coordinates of proteins are obtained from the Protein Data Bank (PDB) and converted into contact maps. Since the PDB contains extensive redundancy in terms of both sequence and structure, identical and similar sequences are removed by `BLASTClust` of at 95% sequence identity. Then, a representative PDB chain that is annotated is selected from each cluster (i.e., has at least one GO term in at least one of the three ontologies) and is of high quality (i.e., has a high-resolution structure). To augment the dataset with specificity and diversity, homology models are also generated for proteins with no PDB structure available using SWISS-MODEL. The contact maps are constructed such that two resides are in contact if the distance between their corresponding C_α atoms is < 10 Å.

3 Robustness and Justification

3.1 Quantitative Analysis

Objectively and rigorously assessing the performance of different protein function prediction methods is important to advance the field. The Critical Assessment of Function Annotation (CAFA) global, community-wide experiment held every few years to blindly assess protein function prediction methods. It uses proteins whose function annotations are not available as targets for participating methods to predict their function.

`DeepFRI` also adopts the commonly used metrics for evaluating GO term predictions, by (1) the protein-centric maximum F-score (F_{max}), and (2) term-centric area under precision-recall (AUPR) curve.

Mathematically, the maximum F-score measures the peak performance of assigning GO terms/EC numbers to a protein, and is computed as a harmonic mean of the precision and recall, given by:

$$F_{\text{max}} = \max_{\tau} F_1(\tau), \text{ where } F_1(\tau) = 2 \times \frac{\text{Precision}(\tau) \times \text{Recall}(\tau)}{\text{Precision}(\tau) + \text{Recall}(\tau)},$$

and $\tau \in [0, 1]$ is a decision threshold for assigning a GO term to a protein if the predicted probability is greater than τ. In contrast, the term-centric AUPR measures the model's ability to balance precision and recall, which is defined as

$$\text{AUPR} = \int_0^1 \text{Precision}(\tau) d\tau.$$

Experiment 1. *How good is* `DeepFRI` *when compared to* `DeepGo`, `BLAST`, *and* `FunFams`*?*

Method. The performance of `DeepFRI` is compared to other state-of-the art and baseline methods. In terms of protein-centric F_{max}, `DeepFRI` outperforms other methods on MF- and BP-GO terms (Fig. 2a,e). `DeepFRI` also learns general structure-function relationships more robustly than other methods by predicting MF-GO terms of proteins with low sequence identity to the training set (Fig. 2b).

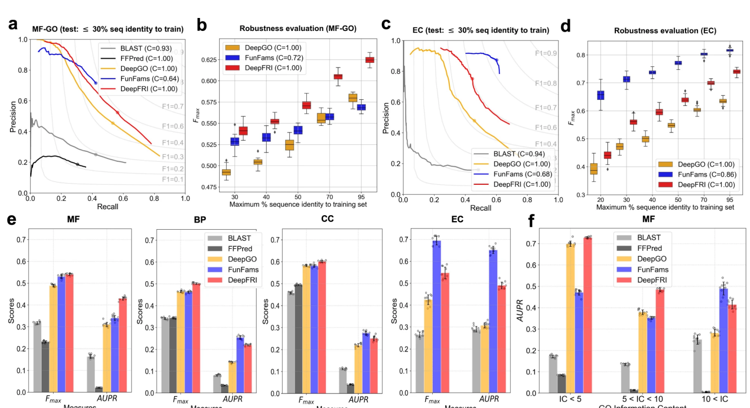

Figure 2: Distribution of protein-centric F_{max} score and function-centric AUPR score summarized over all test proteins in different ontologies and EC numbers, respectively.

3.2 Qualitative Analysis

Experiment 2. *Is* `DeepFRI` *sensitive to the modeling errors in protein structure prediction?*

Method. By investigating the performance of `DeepFRI` trained solely on PDB structures. Experiment is conducted by extracting 700 unseen, experimentally annotated PDB chains with their NATIVE structures, and lowest energy (LE) model by Rosetta and DMPFold. The results show that `DeepFRI` exhibits higher performance (with $F_{\text{max}} = 0.657/0.633/0.619$ for native structures, models from DMPFold, Rosetta, respectively), far better than the CNN-based method `DeepGO` ($F_{\text{max}} = 0.525$ for native structures) even when accounting for errors in predicted contact maps.

The robustness in predicting GO terms with degrading quality of predicted models is further tested (Fig. 3a,d). To do this, a set of Rosetta models with different template modeling scores (TM-scores) against their corresponding native structure is generated. It is observed that even for low TM-scores, `DeepFRI` obtains better performance in GO term classification than the sequence-only CNN-based method. Fig. 3c also shows the output of `DeepFRI` with varying quality (TM-score) of Rosetta models of rat intestinal lipid binding apo protein. For models with TM-scores > 0.58, `DeepFRI` correctly predicts four GO terms including lipid binding, whereas for a TM-score > 0.73, `DeepFRI` correctly predicts even more specific function - fatty acid binding.

Interpretation. The high denoising ability of the GCN is implied by a high correlation between GCN features extracted from NATIVE and LE contact maps. However, the high tolerance for predicting functions from low-quality models justifies the powerful language model features of LSTM-LM, which the model is mainly relying on when making those predictions.

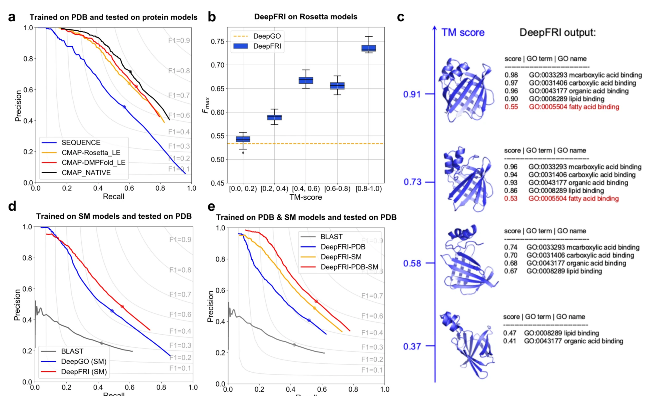

Figure 3: Performance of `DeepFRI` in MF-GO with experimental structures and protein models.

Experiment 3. *How to tackle the imbalanceness in the number of positive and negative examples?*

Method. By investigating the inclusion of homology models into the training data. $\sim$ 220k non-redundant homology models are obtained from the SWISS-MODEL repository, which results in more training examples and larger GO term coverage, and consequently more accurate performance results (Fig. 3e).

Experiment 4. *How to further justify the robustness of* `DeepFRI` *in terms of its ability to leverage structural information?*

Method. By testing `DeepFRI`'s performance to predict residue-level annotations, which can be visualized using post-processed gradient-weighted Class Activation Maps (grad-CAM). It does so by first computing the contribution of each graph convolutional feature map of the model (trained on the MF-GO dataset) to the GO term prediction, and then by summing the feature maps with positive contributions to obtain a final residue-level activation map.

`DeepFRI` detects function-specific structural sites by identifying residues relevant for making accurate GO term prediction (for DeepFRI model trained on MF-GO terms). Figure 4a shows the grad-CAM identified residues for a calcium ion binding of α-parvalbumin protein. The two highest peaks in the profile correspond to the calcium-binding residues in the structure of the protein (Fig. 4a, left). A high area of `DeepFRI` under the ROC curve indicates its high correspondence between annotated binding sites and our predictions, meaning high accuracies in residue-level predictions, even if it is not trained to predict active sites of proteins. However, we stress that for the GO terms related to binding, grad-CAMs do not necessarily identify binding residues/regions; instead, they identify regions that are conserved among the sequences annotated with the same function. This can be explained by Occam's razor principle, where the model prefers to use the simplest explanation for the best possible accuracy.

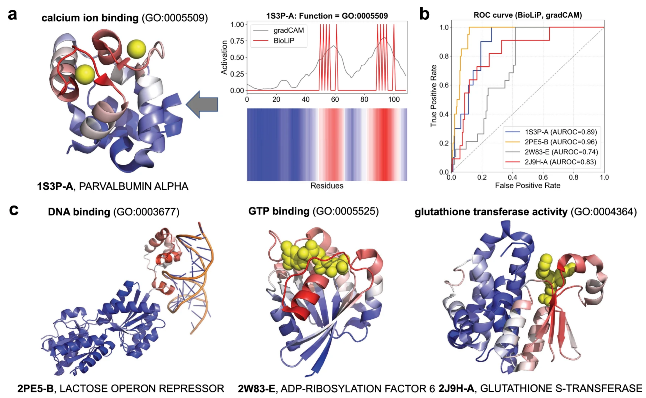

Figure 4: Automatic mapping of function prediction to sites on protein structures.

Experiment 5. *How to evaluate* `DeepFRI`'s *performance in a realistic scenario?*

Method. By evaluating the performance of `DeepFRI` using a temporal holdout strategy. A test set of PDB chains is composed by looking at the difference in GO annotations of the PDB chains in the SIFTS56 database between two releases (separated by $\sim$ 6 months - releases 18 June 2019 and 04 January 2020). It is found that `DeepFRI` significantly outperforms both `BLAST` and `DeepGO`. There are even examples of PDB chains with correctly predicted GO terms for which both `BLAST` and `DeepGO` are failing to produce any meaningful predictions, indicating again the importance of structural information.

4 Conclusion and Future Works

In this report, we have presented `DeepFRI` for protein function prediction, which combines LSTM-LM and GCN to leverage sequence and structure information. The model outperforms state-of-the-art methods in terms of prediction accuracy and generalization to unseen protein structures. We have also discussed different methods that `DeepFRI` has taken to justify its robustness in terms of its ability to predict functions from low-quality protein models and to identify key residues that contribute to protein function, providing means of interpreting prediction results.

We anticipate that future works include the collection of data to extend the coverage of rare protein functions, improvements on the model's architecture and interpretability, and the generalization of the model that takes the quaternary protein structure into consideration.

With ongoing improvements in AI models and datasets, we are confident that the accuracy and applicability of protein function predictions will continue to expand, paving the way for breakthroughs in drug discovery, synthetic biology, and molecular medicine.

References

[1] Gligorijević, V., Renfrew, P.D., Kosciolek, T. *et al.* (2021) Structure-based protein function prediction using graph convolutional networks. *Nature Communication* **12**: 3168. doi:10.1038/s41467-021-23303-9.

[2] Boadu, F., Lee, A., & Cheng, J. (2025) Deep learning methods for protein function prediction. Proteomics **25**, e2300471. doi:10.1002/pmic.202300471.

Appendix

We explore more on the details of the model architecture, and discuss some training tricks, and technical details on the generation of grad-CAMs in this section.

Model Architecture

We use one-hot encoding to represent a protein sequence with L amino acid residues as a feature matrix $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_L] \in \{0,1\}^{L\times c}$, where $c = 26$ one-hot indicators are used to encode each type of amino acid (20 standard, 5 non-standard, and the gap symbol) at position i. The sequences are represented using one-hot encoding. The language model architecture is comprised of two stacked forward LSTM layers with 512 units each, The LSTM-LM model is trained for 5 epochs using an ADAM optimizer with a learning rate 0.001 and a batch size of 128. All hyperparameters are determined through a grid search based on the model's performance on the validation set.

The residue-level features, extracted from the final LSTM layer's hidden states, H^{LM}, are combined with 1-hot representation of sequences, X, through learnable non-linear mapping:

$$H^{\text{input}} = \text{ReLU}(H^{\text{LM}}W^{\text{LM}} + XW^X + B),$$

where H^{input} is the final residue-level feature representation passed to the first GCN layer. Only the parameters, W^{LM}, W^X, and b are trained with the parameters of the GCN, while all the parameters of the LSTM-LM are frozen during the training because it is more efficient.

The GCN takes both the adjacency matrix $A \in \mathbb{R}^{L\times L}$ and the residue-level embeddings from the previous layer, $H^{(l)} \in \mathbb{R}^{L\times c_l}$ and outputs the residue-level embeddings in the next layer, using the plain Kipf and Welling graph convolution $H^{l+1} \in \mathbb{R}^{L\times c_{l+1}}$:

$$H^{l+1} = \text{GCN}(A, H^l) = \text{ReLU}(\tilde{D}^{-0.5}\tilde{A}\tilde{D}^{-0.5}H^{(l)}W^{(l)}),$$

where $\tilde{A} = A + I_L$ is the adjacency matrix with added self-connections, $\tilde{D}$ is the diagonal degree matrix with entries $\tilde{D}_{ii} = \sum_{j=1}^{L}\tilde{A}_{ij}$, and $W^{(l)} \in \mathbb{R}^{c_l\times c_{l+1}}$.

The final protein representation is obtained by first concatenating features from all layers into a single feature matrix, $H = [H^{(1)}, H^{(2)}, \ldots, H^{(L)}] \in \mathbb{R}^{L\times\sum_{l=1}^{L}c_l}$, where $L = 3$ is the number of GCN layers, and then by performing a global pooling layer after which we obtain a fixed vector representation of a protein structure, $h^{\text{pool}} \in \mathbb{R}^{\sum_{l=1}^{L}c_l}$. The global pooling is obtained by a sum operator over L residues: $h^{\text{pool}} = \sum_{l=1}^{L}H_l$. Subsequently, the protein representation is passed through a fully connected layer with a ReLU activation function to compute the hidden representation from the pooled representation.

This is then followed by another fully connected layer which is used for mapping the hidden representation from the previous layer to a $|\text{GO}| \times 2$ output; that is, two activations for each GO term. These activations are transformed by a softmax activation function, outputting the positive and negative probability for each GO term/EC number, i.e., the final layer outputs probability vector $\hat{y}$ of dimension $|\text{GO}| \times 2$ ($|\text{EC}| \times 2$ for EC numbers) for predicting positive and negative probabilities of GO terms (EC numbers).

Measuring Specificity

To evaluate the robustness of `DeepFRI`, we need to differentiate the GO terms based on their specificity, measured by its Shannon information content (IC):

$$\text{IC}(\text{GO}_i) = -\log_2 \text{Prob}(\text{GO}_i),$$

where $\text{Prob}(\text{GO}_i)$ is the probability of observing GO term i in the UniProt-GOA database (n_i/n, where n_i is the number of proteins annotated with GO term i and n is the total number of proteins in UniProt-GOA). Infrequent GO terms, which are considered to be more specific, have higher IC values.

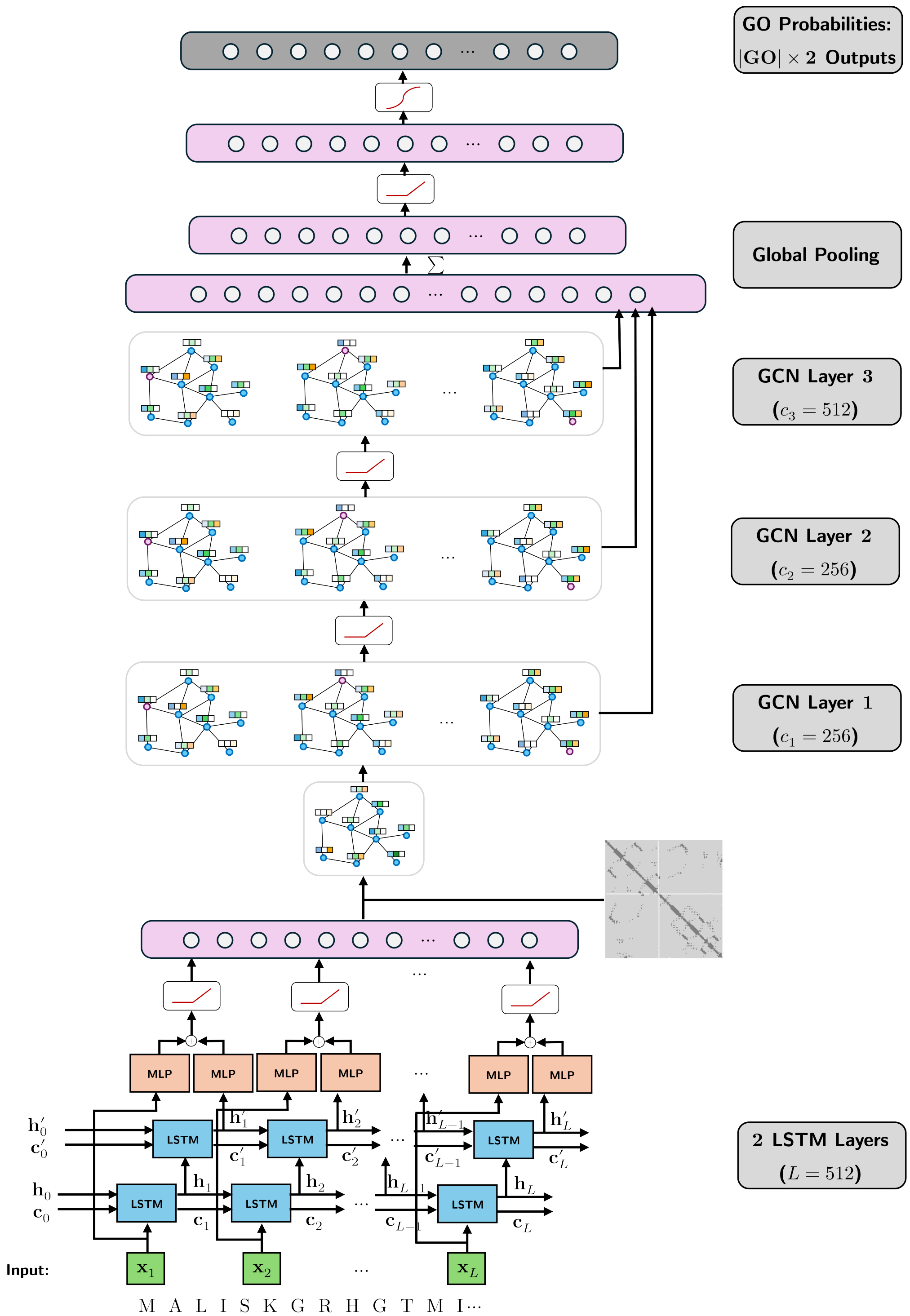

Figure 5: Detailed Schematics of `DeepFRI`.

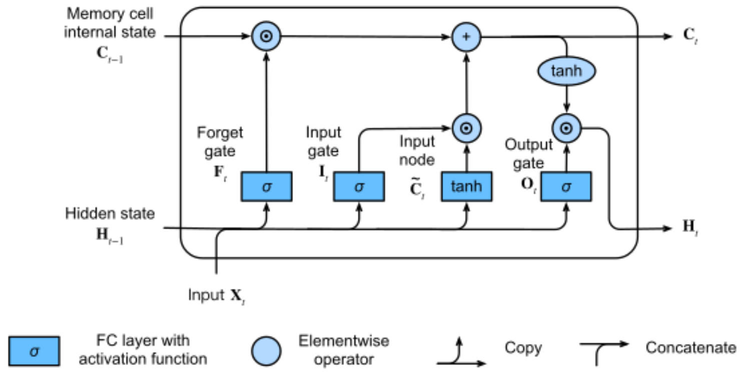

Figure 6: The inner components of an LSTM block.

Training Tricks

To account for imbalanced labels in the dataset, the GCN is trained to minimize the weighted binary cross-entropy cost function that gives higher weights to the GO term with fewer training examples:

$$\mathcal{L}(\Theta) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{|\mathrm{GO}|}\sum_{k=1}^{2} w_j y_{ijk} \log(\hat{y}_{ijk}),$$

where Θ is the set of all parameters in all layers to be learned, $w_j = N/N_j^+$ is the class weight for protein function j, with N_j^+ being the number of positive samples associated with function j, and N is the total number of samples, $y_{ijk}, \hat{y}_{ijk}$ is the true binary indicator and predicted probability for sample i and function j. In the inference phase, GO terms/EC numbers are assigned to the protein sequence if the positive probability is > 0.5.

All hyperparameters are determined through a grid search based on the model's performance on the validation set. The validation set is comprised of $\sim 10\%$ randomly chosen samples from the training set. To avoid overfitting, an early stopping criterion is utilized to stop training if the validation loss does not improve in 5 epochs. We use the ADAM optimizer with a learning rate $lr = 0.0001$, $\beta_1 = 0.95$, and $\beta_2 = 0.95$ and a batch size of 64. The default number of epochs is 200.

Localization of Residual-Level Annotations by grad-CAM

In grad-CAM, the contribution of each filter, k, in the last convolutional layer is computed, by taking the derivative of the output of the prediction for the protein function l, y^l, with respect to feature map $\mathbf{F}_k \in \mathbb{R}^L$ over the whole sequence of length L:

$$w_k^l = \sum_{i=1}^{L} \frac{\partial y^l}{\partial F_{k,i}},$$

where w_k^l represents the importance of feature map k for predicting function l, obtained by summing the contribution from each individual residue. Finally, we obtain the function-specific heatmap in a residue space by making the weighted sum over all feature maps in the last convolutional layer:

$$\mathrm{CAM}^l[i] = \mathrm{ReLU}\left(\sum_k w_k^l F_{k,l}\right),$$

where the ReLU function ensures that only features with positive influence on the functional label are preserved; $\mathrm{CAM}^l[i]$ indicates the relative importance of residue i to function l. The advantage of grad-CAM is that it does not require re-training or changes in the architecture of the model which makes it computationally efficient and directly applicable to the model.